INTERPROCESS COMMUNICATION AND SYNCHRONIZATION

Inter process communication (IPC) is used for exchanging data between multiple threads in one or more processes or programs. The Processes may be running on single or multiple computers connected by a network. The full form of IPC is Inter-process communication.

It is a set of programming interface which allow a programmer to coordinate activities among various program processes which can run concurrently in an operating system. This allows a specific program to handle many user requests at the same time.

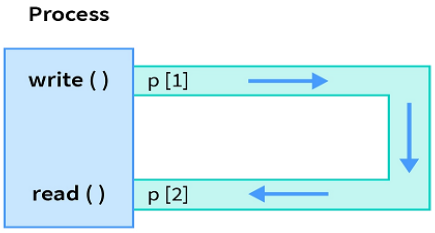
Approaches for Inter-Process Communication

Here, are few important methods for interprocess communication:

Inter-Process Communication Approaches

Pipes

Pipe is widely used for communication between two related processes. This is a half-duplex method, so the first process communicates with the second process. However, in order to achieve a full-duplex, another pipe is needed.

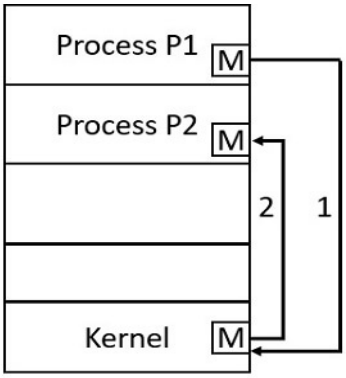


Message Passing:

It is a mechanism for a process to communicate and synchronize. Using message passing, the process communicates with each other without resorting to shared variables.

IPC mechanism provides two operations:

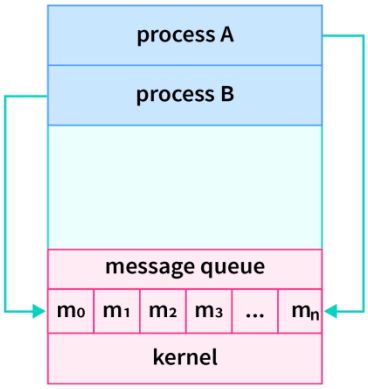
- Send (message)- message size fixed or variable
- Received (message)



Message Passing System

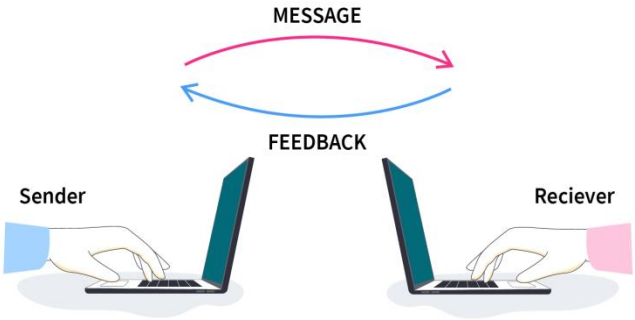
Message Queues:

A message queue is a linked list of messages stored within the kernel. It is identified by a message queue identifier. This method offers communication between single or multiple processes with full-duplex capacity.



Direct Communication:

In this type of inter-process communication process, should name each other explicitly. In this method, a link is established between one pair of communicating processes, and between each pair, only one link exists.

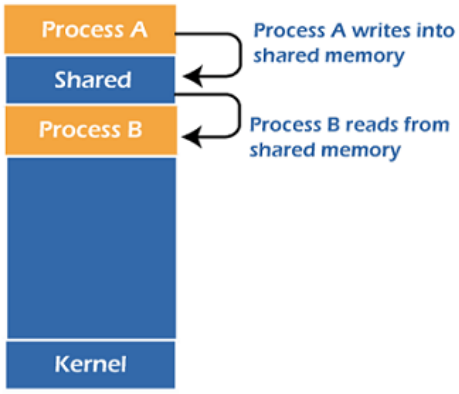


Indirect Communication:

Indirect communication establishes like only when processes share a common mailbox each pair of processes sharing several communication links. A link can communicate with many processes. The link may be bi-directional or unidirectional.

Shared Memory:

Shared memory is a memory shared between two or more processes that are established using shared memory between all the processes. This type of memory requires to protected from each other by synchronizing access across all the processes.



FIFO:

Communication between two unrelated processes. It is a full-duplex method, which means that the first process can communicate with the second process, and the opposite can also happen.

Why IPC?

Here, are the reasons for using the interprocess communication protocol for information sharing:

- It helps to speedup modularity
- Computational
- Privilege separation
- Convenience

- Helps operating system to communicate with each other and synchronize their actions.

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PROCESS SYNCHRONIZATION

When multiple processes execute concurrently sharing system resources, then inconsistent results might be produced.

- Process Synchronization is a mechanism that deals with the synchronization of processes.
- It controls the execution of processes running concurrently to ensure that consistent results are produced.

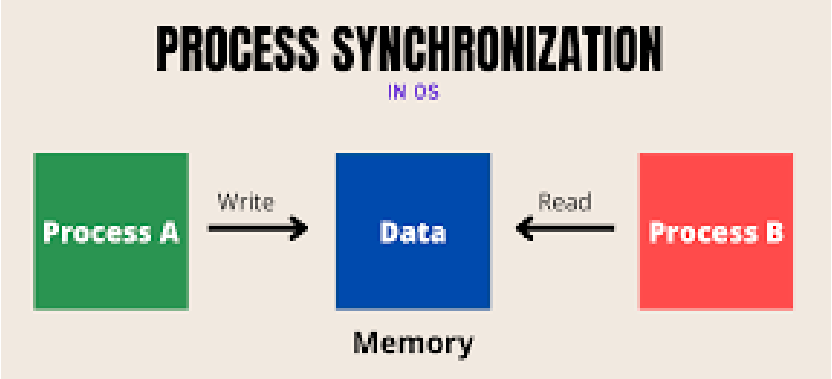
Need of Synchronization-

Process synchronization is needed-

- When multiple processes execute concurrently sharing some system resources.
- To avoid the inconsistent results.

How Process Synchronization Works?

For Example, process A changing the data in a memory location while another process B is trying to read the data from the **same** memory location. There is a high probability that data read by the second process will be erroneous.



On the basis of synchronization, processes are categorized as one of the following two types:

- **Independent Process**: The execution of one process does not affect the execution of other processes.
- **Cooperative Process**: A process that can affect or be affected by other processes executing in the system.

Sections of a Program

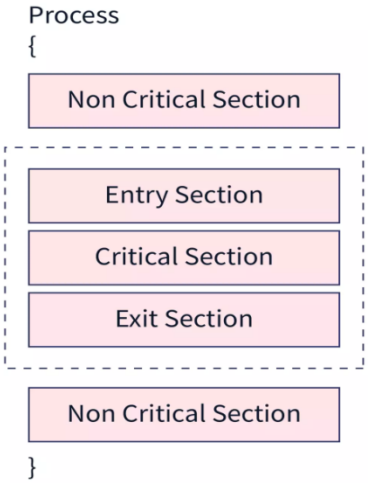
Here, are four essential elements of the critical section:

- **Entry Section:** It is part of the process which decides the entry of a particular process.
- **Critical Section:** This part allows one process to enter and modify the shared variable.

- **Exit Section:** Exit section allows the other process that are waiting in the Entry Section, to enter into the Critical Sections. It also checks that a process that finished its execution should be removed through this Section.
- **Remainder Section:** All other parts of the Code, which is not in Critical, Entry, and Exit Section, are known as the Remainder Section.

CRITICAL SECTION

- Critical Section refers to the segment of code or the program which tries to access or modify the value of the variables in a shared resource.
- The section above the critical section is called the **Entry Section**. The process that is entering the critical section must pass the entry section.
- The section below the critical section is called the **Exit Section**.
- The section below the exit section is called the **Reminder Section** and this section has the remaining code that is left after execution.



What is the Critical Section Problem in OS?

When there is more than one process accessing or modifying a shared resource at the same time, then the value of that resource will be determined by the last process. This is called the **race condition**.

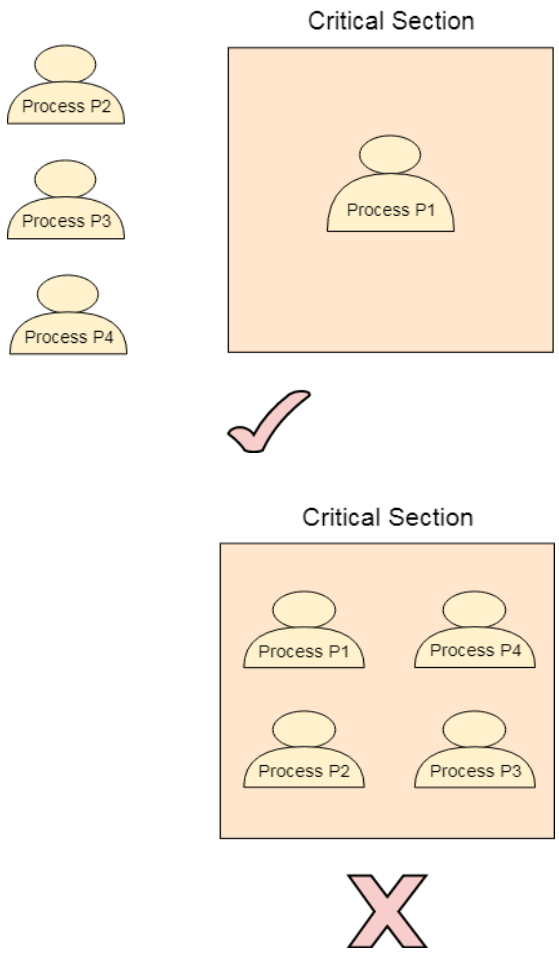
The **critical section problem** is to make sure that only one process should be in a critical section at a time. When a process is in the critical section, no other processes are allowed to enter the critical section. This solves the race condition.

Solutions to the Critical Section Problem

Primary

1. **Mutual Exclusion**

Our solution must provide mutual exclusion. By Mutual Exclusion, we mean that if one process is executing inside critical section then the other process must not enter in the critical section.



2. **Progress**

   Progress means that if one process doesn't need to execute into critical section then it should not stop other processes to get into the critical section.

- If a process is in the critical section and another process arrives, then the new process must wait until the first process exits the critical section. In such cases, the process that is waiting to enter the critical section should not wait for an unlimited period. This is called **progress**.

Secondary

1. **Bounded Waiting**

   We should be able to predict the waiting time for every process to get into the critical section. The process must not be endlessly waiting for getting into the critical section.

- If a process wants to enter into the critical section, then there should be a specified time that the process can be made to wait. This property is called **bounded waiting**.

2. **Architectural Neutrality**

   Our mechanism must be architectural natural. It means that if our solution is working fine on one architecture then it should also run on the other ones as well.

SEMAPHORE

- A semaphore is a simple integer variable used to provide synchronization among the processes.
- Semaphore is essentially a non negative integer that is used to solve the critical section problem by acting as a signal. It is a concept in operating systems for the synchronization of concurrent processes.

Semaphore was proposed by Dijkstra in 1965 which is a very significant technique to manage concurrent processes by using a simple integer value, which is known as a semaphore. A semaphore is simply an integer variable that is shared between threads. This variable is used to solve the critical section problem and to achieve process synchronization in the multiprocessing environment.

Characteristic of Semaphore

Here, are characteristic of a semaphore:

- It is a mechanism that can be used to provide synchronization of tasks.
- It is a low-level synchronization mechanism.
- Semaphore will always hold a non-negative integer value.
- Semaphore can be implemented using test operations and interrupts, which should be executed using file descriptors.

Semaphores are integer variables that are used to solve the critical section problem by using two atomic operations, wait and signal that are used for process synchronization.

The definitions of wait and signal are as follows −

- **Wait**
  The wait operation decrements the value of its argument S, if it is positive. If S is negative or zero, then no operation is performed.

```
wait(S)
{
  while (S<=0);


  S--;
}
```

- **Signal**
  The signal operation increments the value of its argument S.

```
signal(S)
```

```
{
  S++;
}
```

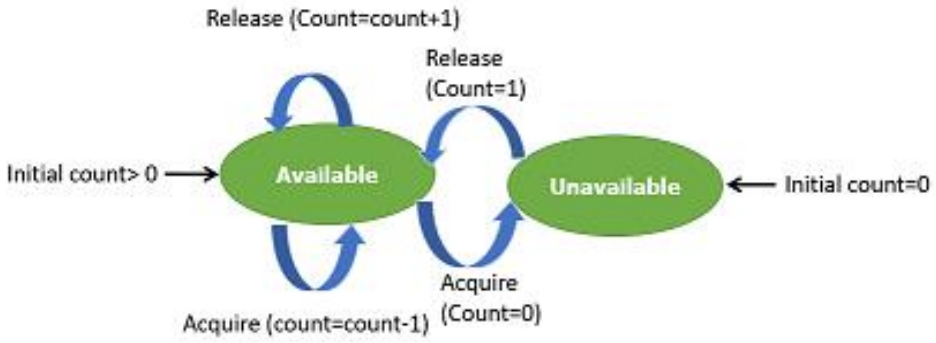
Types of Semaphores

The two common kinds of semaphores are

- Counting semaphores
- Binary semaphores.

Counting Semaphores

This type of Semaphore uses a count that helps task to be acquired or released numerous times. If the initial count = 0, the counting semaphore should be created in the unavailable state.



Example(A)

```
Down(semaphore S)
{
        S.value=S.value-1
        If(S.value<=0)

            {
                Select a process from suspend list wake up();
            }
}
```

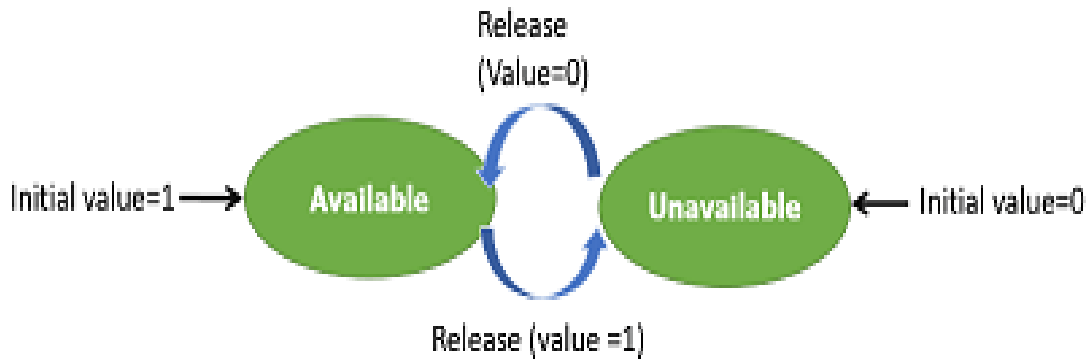
Example(B)

```
up(semaphore S)
{
        S.value=S.value+1
```

```
        If(S.value<0)

            {
                Put process (pcb) in suspend list Sleep();
            }
        Else
                Return;
}
```

Binary Semaphores

The binary semaphores are quite similar to counting semaphores, but their value is restricted to 0 and 1. In this type of semaphore, the wait operation works only if semaphore = 1, and the signal operation succeeds when semaphore= 0. It is easy to implement than counting semaphores.



Example(A)

```
Down(semaphore S)
{
        If(S.value==1)
            {
                S.value=0;
            }
        Else
             {
                Block the process and place in suspend list, sleep();
            }
}
```

Example(B)

```
up(semaphore S)
{
```

```
        If(suspend list is empty)
            {
                S.value=1;
            }
        Else
             {
                select the process from the suspend list and wake up();
            }
}
```

Counting Semaphore vs. Binary Semaphore

Here, are some major differences between counting and binary semaphore:

Counting Semaphore	**Binary Semaphore**
No mutual exclusion	Mutual exclusion
Any integer value	Value only 0 and 1
More than one slot	Only one slot
Provide a set of Processes	It has a mutual exclusion mechanism.

Advantages of Semaphores

Here, are pros/benefits of using Semaphore:

- It allows more than one thread to access the critical section
- Semaphores are machine-independent.
- Semaphores are implemented in the machine-independent code of the microkernel.
- They do not allow multiple processes to enter the critical section.
- As there is busy waiting in semaphore, there is never a wastage of process time and resources.
- They are machine-independent, which should be run in the machine-independent code of the microkernel.

Disadvantage of semaphores

Here, are cons/drawback of semaphore

- One of the biggest limitations of a semaphore is priority inversion.
- The operating system has to keep track of all calls to wait and signal semaphore.
- Their use is never enforced, but it is by convention only.
- In order to avoid deadlocks in semaphore, the Wait and Signal operations require to be executed in the correct order.
- Semaphore programming is a complicated, so there are chances of not achieving mutual exclusion.
- It is also not a practical method for large scale use as their use leads to loss of modularity.

CLASSICAL PROBLEMS IN CONCURRENT PROGRAMMING

The classical problems of synchronization are as follows:

1. Bound-Buffer problem
2. Sleeping barber problem
3. Dining Philosophers problem
4. Readers and writers problem

Bounded Buffer Problem

Because the buffer pool has a maximum size, this problem is often called the **Bounded buffer problem**.

- This problem is generalised in terms of the **Producer Consumer problem**, where a **finite** buffer pool is used to exchange messages between producer and consumer processes.
- Solution to this problem is, creating two counting semaphores "full" and "empty" to keep track of the current number of full and empty buffers respectively.
- In this Producers mainly produces a product and consumers consume the product, but both can use of one of the containers each time.
- The main complexity of this problem is that we must have to maintain the count for both empty and full containers that are available.

One solution to this problem is to use semaphores. The semaphores which will be used here are:

- m, a **binary semaphore** that is used to acquire and release the lock.
- empty, a **counting semaphore** whose initial value is the number of slots in the buffer, since, initially all slots are empty.
- full, a **counting semaphore** whose initial value is 0.

The Producer Operation

The pseudocode of the producer function looks like this:

```
do
{
    // wait until empty > 0 and then decrement 'empty'
    wait(empty);
    // acquire lock
    wait(mutex);

    /* perform the insert operation in a slot */

    // release lock
    signal(mutex);
    // increment 'full'
    signal(full);
}
while(TRUE)
```

- Looking at the above code for a producer, we can see that a producer first waits until there is at least one empty slot.
- Then it decrements the **empty** semaphore because there will now be one less empty slot since the producer is going to insert data in one of those slots.
- Then, it acquires a lock on the buffer, so that the consumer cannot access the buffer until the producer completes its operation.
- After performing the insert operation, the lock is released and the value of **full** is incremented because the producer has just filled a slot in the buffer.

The Consumer Operation

The pseudocode for the consumer function looks like this:

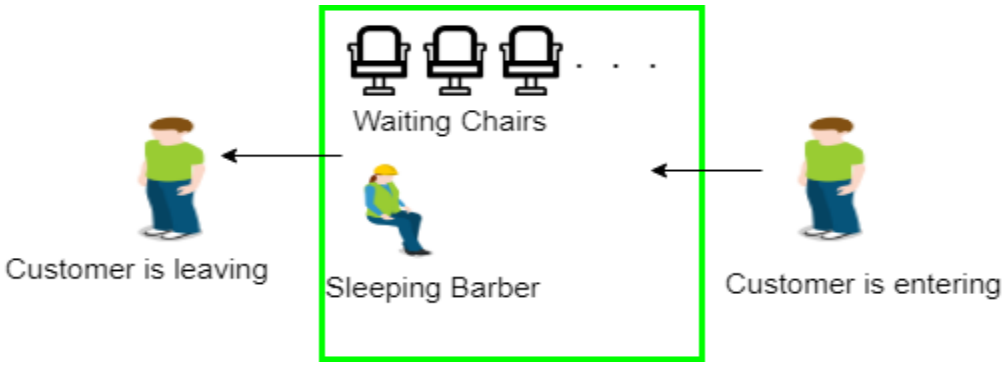
```
do
{
    // wait until full > 0 and then decrement 'full'
    wait(full);
    // acquire the lock
    wait(mutex);

    /* perform the remove operation in a slot */

    // release the lock
    signal(mutex);
    // increment 'empty'
    signal(empty);
}
while(TRUE);
```

- The consumer waits until there is at least one full slot in the buffer.
- Then it decrements the **full** semaphore because the number of occupied slots will be decreased by one after the consumer completes its operation.
- After that, the consumer acquires a lock on the buffer.
- Following that, the consumer completes the removal operation so that the data from one of the full slots are removed.
- Then, the consumer releases the lock.

- Finally, the **empty** semaphore is incremented by 1, because the consumer has just removed data from an occupied slot, thus making it empty.

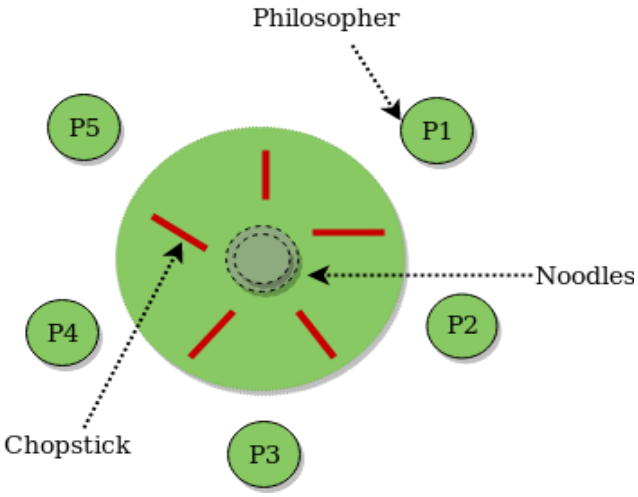
SLEEPING BARBER PROBLEM:

Barber shop with one barber, one barber chair and N chairs to wait in. When no customers the barber goes to sleep in barber chair and must be woken when a customer comes in. When barber is cutting hair new customers take empty seats to wait, or leave if no vacancy.



DINING PHILOSOPHERS PROBLEM

- The dining philosopher's problem involves the allocation of limited resources to a group of processes in a deadlock-free and starvation-free manner.
- There are five philosophers sitting around a table, in which there are five chopsticks/forks kept beside them and a bowl of rice in the centre, When a philosopher wants to eat, he uses two chopsticks - one from their left and one from their right. When a philosopher wants to think, he keeps down both chopsticks at their original place.



Solution

From the problem statement, it is clear that a philosopher can think for an indefinite amount of time. But when a philosopher starts eating, he has to stop at some point in time. The philosopher is in an endless cycle of thinking and eating.

An array of five semaphores, stick[5]for each of the five chopsticks.

The code for each philosopher looks like this:

```
while(TRUE)
{
    wait(stick[i]);
    /*
        mod is used because if i=5, next
        chopstick is 1 (dining table is circular)
    */
    wait(stick[(i+1) % 5]);

    /* eat */
    signal(stick[i]);

    signal(stick[(i+1) % 5]);
    /* think */
}
```

When a philosopher wants to eat the rice, he will wait for the chopstick at his left and picks up that chopstick. Then he waits for the right chopstick to be available, and then picks it too. After eating, he puts both the chopsticks down.

But if all five philosophers are hungry simultaneously, and each of them picks up one chopstick, then a deadlock situation occurs because they will be waiting for another chopstick forever. The possible solutions for this are:

- There should be at most four philosophers on the table.
- An even philosopher should pick the right chopstick and then the left chopstick while an odd philosopher should pick the left chopstick and then the right chopstick.
- A philosopher should only be allowed to pick their chopstick if both are available at the same time.

THE READERS WRITERS PROBLEM

- In this problem there are some processes(called **readers**) that only read the shared data, and never change it, and there are other processes(called **writers**) who may change the data in addition to reading, or instead of reading it.
- There are various type of readers-writers problem, most centred on relative priorities of readers and writers.
- The main complexity with this problem occurs from allowing more than one reader to access the data at the same time.

Suppose that a database is to be shared among several concurrent processes. Some of these processes may want only to read the database, whereas others may want to update (that is, to read and write) the database. We distinguish between these two types of processes by referring to the former as readers and to the latter as writers. Precisely in OS we call this situation as the readers-writers problem. Problem parameters:

- One set of data is shared among a number of processes.
- Once a writer is ready, it performs its write. Only one writer may write at a time.
- If a process is writing, no other process can read it.
- If at least one reader is reading, no other process can write.
- Readers may not write and only read.

The Solution

From the above problem statement, it is evident that readers have higher priority than writer. If a writer wants to write to the resource, it must wait until there are no readers currently accessing that resource.

Here, we use one **mutex** m and a **semaphore** w. An integer variable read_count is used to maintain the number of readers currently accessing the resource. The variable read_count is initialized to 0. A value of 1 is given initially to m and w.

Instead of having the process to acquire a lock on the shared resource, we use the mutex m to make the process to acquire and release lock whenever it is updating the read_count variable.

The code for the **writer** process looks like this:

```
while(TRUE)
{
    wait(w);

   /* perform the write operation */

   signal(w);
}
```

And, the code for the **reader** process looks like this:

```
while(TRUE)
{
    //acquire lock
    wait(m);
    read_count++;
    if(read_count == 1)
        wait(w);

    //release lock
    signal(m);

    /* perform the reading operation */

    // acquire lock
    wait(m);
    read_count--;
    if(read_count == 0)
        signal(w);

    // release lock
    signal(m);
}
```

Here is the Code uncoded(explained)

- As seen above in the code for the writer, the writer just waits on the **w** semaphore until it gets a chance to write to the resource.
- After performing the write operation, it increments **w** so that the next writer can access the resource.
- On the other hand, in the code for the reader, the lock is acquired whenever the **read_count** is updated by a process.
- When a reader wants to access the resource, first it increments the **read_count** value, then accesses the resource, and then decrements the **read_count** value.
- The semaphore **w** is used by the first reader which enters the critical section and the last reader who exits the critical section.
- The reason for this is, when the first readers enter the critical section, the writer is blocked from the resource. Only new readers can access the resource now.
- Similarly, when the last reader exits the critical section, it signals the writer using the **w** semaphore because there are zero readers now and a writer can have the chance to access the resource.